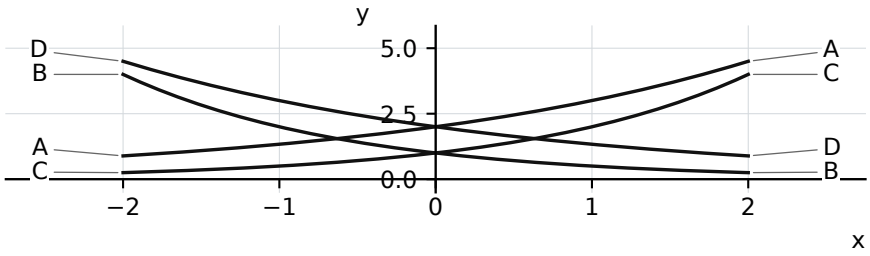
MATH 101 **71852**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 1(2)^x$ $(b)\quad 2(\frac{3}{2})^x$ $(c)\quad 2(\frac{2}{3})^x$ $(d)\quad 1(\frac{1}{2})^x$  	1. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts
2. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.2)+\log(500)$ $(b)\quad \frac{\log_3(8)}{\log_3(4)}$	2. (a) 10 pts (b) 10 pts
3. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=3$ $x-y=1$ $(b)\quad x^2+y^2=5$ $y=x+1$ $(c)\quad x^2+y^2\le 5,\ y>x+1$	3. (a) 6 pts (b) 10 pts (c) 4 pts

Test continues on the other side of the sheet.

MATH 101 **71852**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x-1) + 1$	4. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
5. Solve both equations exactly. $(a)\quad 4^{x-1} = 2^{x+2}$ $(b)\quad \log_2(x-1) + \log_2(3) = 2$	5. (a) 8 pts (b) 12 pts

71852

TEST PAGE 1

1. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

2. (a)

2

10 pts

(b)

$\frac{3}{2}$

10 pts

3. (a)

$(2, 1)$

6 pts

(b)

$(1, 2); (-2, -1)$

10 pts

(c)

No

4 pts

TEST PAGE 2

4. (a)

$(1, \infty)$

4 pts

(b)

$(-\infty, \infty)$

4 pts

(c)

$x = 1$

3 pts

(d)

$(\frac{3}{2}, 0)$

3 pts

(e)

$x \to 1^+ : f(x) \to -\infty$

3 pts

(f)

$x \to \infty : f(x) \to \infty$

3 pts

5. (a)

$x = 4$

8 pts

(b)

$x = \frac{7}{3}$

12 pts

MATH 101 **90470**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-1) + 1$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
2. Solve both equations exactly. $(a)\quad 4^{x-1} = 2^{x+2}$ $(b)\quad \log_2(x-1) + \log_2(3) = 2$	2. (a) 8 pts (b) 12 pts
3. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x + y = 3$ $x - y = 1$ $(b)\quad x^2 + y^2 = 5$ $y = x + 1$ $(c)\quad x^2 + y^2 \le 5,\ y > x + 1$	3. (a) 6 pts (b) 10 pts (c) 4 pts

Test continues on the other side of the sheet.

MATH 101 **90470**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

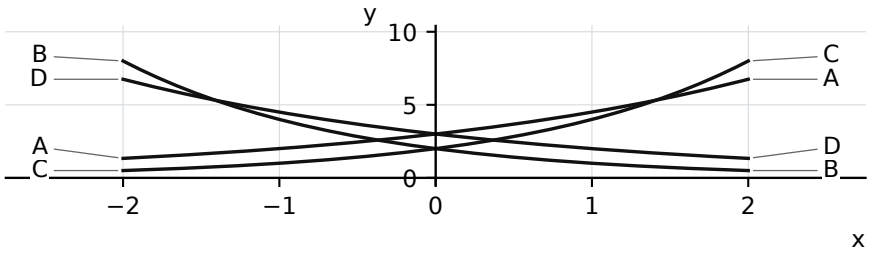
4. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $2(2)^x$

(b) $3(\frac{3}{2})^x$

(c) $3(\frac{2}{3})^x$

(d) $2(\frac{1}{2})^x$



4. (a) 5 pts

(b) 5 pts

(c) 5 pts

(d) 5 pts

5. Evaluate exactly. Here log means base 10.

(a) $\log(0.4) + \log(2500)$

(b) $\frac{\log_3(8)}{\log_3(4)}$

5. (a) 10 pts

(b) 10 pts

90470

TEST PAGE 1

1. (a)
$(1, \infty)$ 4 pts

(b)
$(-\infty, \infty)$ 4 pts

(c)
$x = 1$ 3 pts

(d)
$(\frac{4}{3}, 0)$ 3 pts

(e)
$x \to 1^+ : f(x) \to -\infty$ 3 pts

(f)
$x \to \infty : f(x) \to \infty$ 3 pts

2. (a)
$x = 4$ 8 pts

(b)
$x = \frac{7}{3}$ 12 pts

3. (a)
$(2, 1)$ 6 pts

(b)
$(1, 2); (-2, -1)$ 10 pts

(c)
No 4 pts

TEST PAGE 2

4. (a)
C 5 pts

(b)
A 5 pts

(c)
D 5 pts

(d)
B 5 pts

5. (a)
3 10 pts

(b)
$\frac{3}{2}$ 10 pts

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

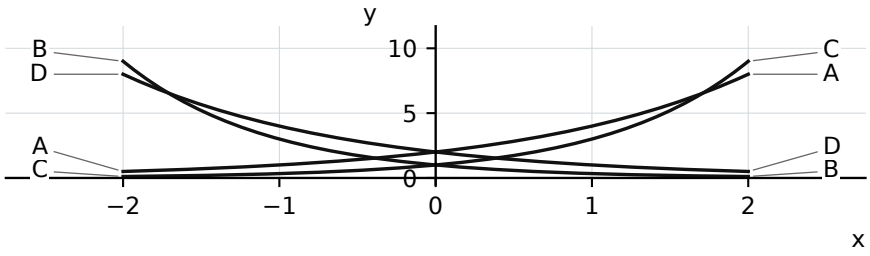
1. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $1(3)^x$

(b) $2(2)^x$

(c) $2(\frac{1}{2})^x$

(d) $1(\frac{1}{3})^x$



1. (a) 5 pts

(b) 5 pts

(c) 5 pts

(d) 5 pts

2. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_4(x-1)+1$

2. (a) 4 pts

(b) 4 pts

(c) 3 pts

(d) 3 pts

(e) 3 pts

(f) 3 pts

3. Solve both equations exactly.

(a) $4^{x-1} = 2^{x+2}$

(b) $\log_2(x-1)+\log_2(3)=2$

3. (a) 8 pts

(b) 12 pts

Test continues on the other side of the sheet.

MATH 101 **79773**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.5)+\log(20000)$ $(b)\quad \frac{\log_3(8)}{\log_3(4)}$	4. (a) 10 pts (b) 10 pts
5. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=3$ $x-y=1$ $(b)\quad x^2+y^2=5$ $y=x+1$ $(c)\quad x^2+y^2\le 5,\ y>x+1$	5. (a) 6 pts (b) 10 pts (c) 4 pts

79773

TEST PAGE 1

1. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

2. (a)

$(1, \infty)$

4 pts

(b)

$(-\infty, \infty)$

4 pts

(c)

$x = 1$

3 pts

(d)

$(\frac{5}{4}, 0)$

3 pts

(e)

$x \to 1^+ : f(x) \to -\infty$

3 pts

(f)

$x \to \infty : f(x) \to \infty$

3 pts

3. (a)

$x = 4$

8 pts

(b)

$x = \frac{7}{3}$

12 pts

TEST PAGE 2

4. (a)

4

10 pts

(b)

$\frac{3}{2}$

10 pts

5. (a)

$(2, 1)$

6 pts

(b)

$(1, 2); (-2, -1)$

10 pts

(c)

No

4 pts

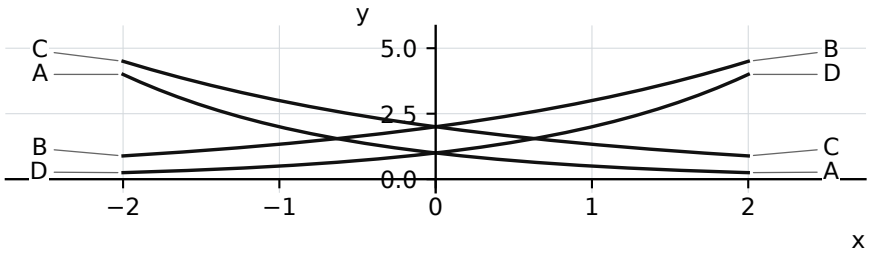
MATH 101 **16256**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(\frac{2}{3})^x$ $(b)\quad 1(\frac{1}{2})^x$ $(c)\quad 1(2)^x$ $(d)\quad 2(\frac{3}{2})^x$ 	1. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts
2. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=5$ $x-y=1$ $(b)\quad y=x^2$ $y=x+2$ $(c)\quad y\geq x^2,\ y<x+2$	2. (a) 6 pts (b) 10 pts (c) 4 pts
3. Solve both equations exactly. $(a)\quad 4^{x-2}=2^{x+3}$ $(b)\quad \log_3(x-2)+\log_3(4)=2$	3. (a) 8 pts (b) 12 pts

Test continues on the other side of the sheet.

MATH 101 **16256**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x - 2) + 2$	4. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
5. Evaluate exactly. Here log means base 10. (a) $\log(0.2) + \log(500)$ (b) $\frac{\log_4(27)}{\log_4(9)}$	5. (a) 10 pts (b) 10 pts

16256

TEST PAGE 1

1. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

2. (a)

$(3, 2)$

6 pts

(b)

$(-1, 1); (2, 4)$

10 pts

(c)

Yes

4 pts

3. (a)

$x = 7$ 8 pts

(b)

$x = \frac{17}{4}$ 12 pts

TEST PAGE 2

4. (a)

$(2, \infty)$ 4 pts

(b)

$(-\infty, \infty)$ 4 pts

(c)

$x = 2$ 3 pts

(d)

$(\frac{9}{4}, 0)$ 3 pts

(e)

$x \to 2^+ : f(x) \to -\infty$ 3 pts

(f)

$x \to \infty : f(x) \to \infty$ 3 pts

5. (a)

2 10 pts

(b)

$\frac{3}{2}$ 10 pts

MATH 101 **67985**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

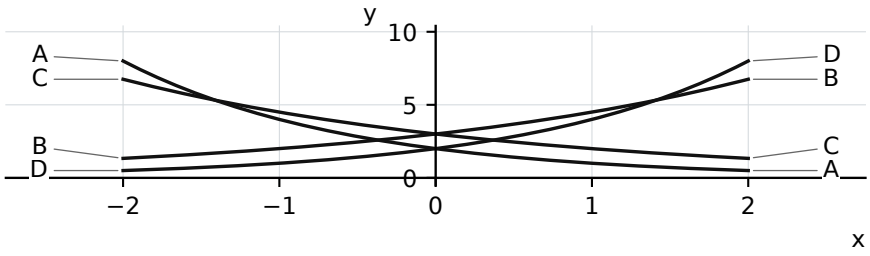
1. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $3(\frac{2}{3})^x$

(b) $2(\frac{1}{2})^x$

(c) $2(2)^x$

(d) $3(\frac{3}{2})^x$



1. (a) 5 pts

(b) 5 pts

(c) 5 pts

(d) 5 pts

2. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_3(x-2)+2$

2. (a) 4 pts

(b) 4 pts

(c) 3 pts

(d) 3 pts

(e) 3 pts

(f) 3 pts

3. Solve both equations exactly.

(a) $4^{x-2} = 2^{x+3}$

(b) $\log_3(x-2)+\log_3(4) = 2$

3. (a) 8 pts

(b) 12 pts

Test continues on the other side of the sheet.

MATH 101 **67985**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=5$ $x-y=1$ $(b)\quad y=x^2$ $y=x+2$ $(c)\quad y\geq x^2,\ y<x+2$	4. (a) 6 pts (b) 10 pts (c) 4 pts
5. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.4)+\log(2500)$ $(b)\quad \frac{\log_4(27)}{\log_4(9)}$	5. (a) 10 pts (b) 10 pts

67985

TEST PAGE 1

1. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

2. (a)

$(2, \infty)$

4 pts

(b)

$(-\infty, \infty)$

4 pts

(c)

$x = 2$

3 pts

(d)

$(\frac{19}{9}, 0)$

3 pts

(e)

$x \to 2^+ : f(x) \to -\infty$

3 pts

(f)

$x \to \infty : f(x) \to \infty$

3 pts

3. (a)

$x = 7$

8 pts

(b)

$x = \frac{17}{4}$

12 pts

TEST PAGE 2

4. (a)

$(3, 2)$

6 pts

(b)

$(-1, 1); (2, 4)$

10 pts

(c)

Yes

4 pts

5. (a)

3

10 pts

(b)

$\frac{3}{2}$

10 pts

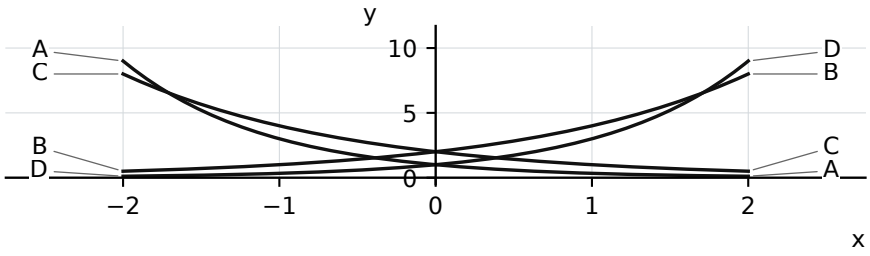
MATH 101 **10118**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x + y = 5$ $x - y = 1$ $(b)\quad y = x^2$ $y = x + 2$ $(c)\quad y \geq x^2,\ y < x + 2$	1. (a) 6 pts (b) 10 pts (c) 4 pts
2. Solve both equations exactly. $(a)\quad 4^{x-2} = 2^{x+3}$ $(b)\quad \log_3(x-2) + \log_3(4) = 2$	2. (a) 8 pts (b) 12 pts
3. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(\frac{1}{2})^x$ $(b)\quad 1(\frac{1}{3})^x$ $(c)\quad 1(3)^x$ $(d)\quad 2(2)^x$ 	3. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts

Test continues on the other side of the sheet.

MATH 101 **10118**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-2) + 2$	4. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
5. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.5) + \log(20000)$ $(b)\quad \frac{\log_4(27)}{\log_4(9)}$	5. (a) 10 pts (b) 10 pts

10118

TEST PAGE 1

1. (a)

$(3, 2)$

6 pts

(b)

$(-1, 1); (2, 4)$

10 pts

(c)

Yes

4 pts

2. (a)

$x = 7$

8 pts

(b)

$x = \frac{17}{4}$

12 pts

3. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

TEST PAGE 2

4. (a)

$(2, \infty)$

4 pts

(b)

$(-\infty, \infty)$

4 pts

(c)

$x = 2$

3 pts

(d)

$(\frac{33}{16}, 0)$

3 pts

(e)

$x \to 2^+ : f(x) \to -\infty$

3 pts

(f)

$x \to \infty : f(x) \to \infty$

3 pts

5. (a)

4

10 pts

(b)

$\frac{3}{2}$

10 pts

MATH 101 **46739**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve both equations exactly.

(a) $4^{x-3} = 2^{x+4}$

(b) $\log_4(x-3) + \log_4(5) = 2$

1. (a) 8 pts

(b) 12 pts

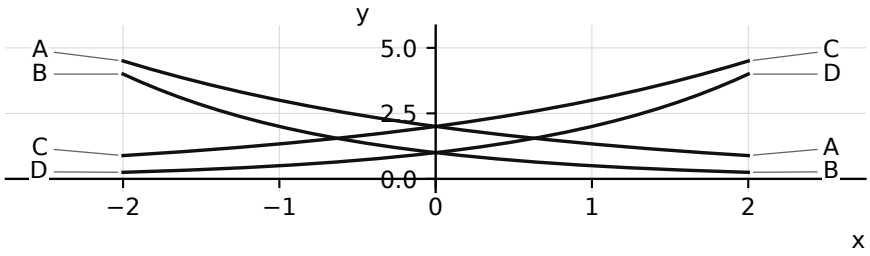
2. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $2(\frac{3}{2})^x$

(b) $2(\frac{2}{3})^x$

(c) $1(2)^x$

(d) $1(\frac{1}{2})^x$



2. (a) 5 pts

(b) 5 pts

(c) 5 pts

(d) 5 pts

3. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_2(x-3) + 3$

3. (a) 4 pts

(b) 4 pts

(c) 3 pts

(d) 3 pts

(e) 3 pts

(f) 3 pts

Test continues on the other side of the sheet.

MATH 101 **46739**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.2)+\log(500)$ $(b)\quad \frac{\log_5(64)}{\log_5(16)}$	4. (a) 10 pts (b) 10 pts
5. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=7$ $x-y=1$ $(b)\quad x^2+y^2=17$ $y=x+3$ $(c)\quad x^2+y^2\le 17,\ y>x+3$	5. (a) 6 pts (b) 10 pts (c) 4 pts

46739

TEST PAGE 1

1. (a)

$x = 10$ — 8 pts

(b)

$x = \frac{31}{5}$ — 12 pts

2. (a)

C — 5 pts

(b)

A — 5 pts

(c)

D — 5 pts

(d)

B — 5 pts

3. (a)

$(3, \infty)$ — 4 pts

(b)

$(-\infty, \infty)$ — 4 pts

(c)

$x = 3$ — 3 pts

(d)

$(\frac{25}{8}, 0)$ — 3 pts

(e)

$x \to 3^+ : f(x) \to -\infty$ — 3 pts

(f)

$x \to \infty : f(x) \to \infty$ — 3 pts

TEST PAGE 2

4. (a)

2 — 10 pts

(b)

$\frac{3}{2}$ — 10 pts

5. (a)

$(4, 3)$ — 6 pts

(b)

$(1, 4); (-4, -1)$ — 10 pts

(c)

No — 4 pts

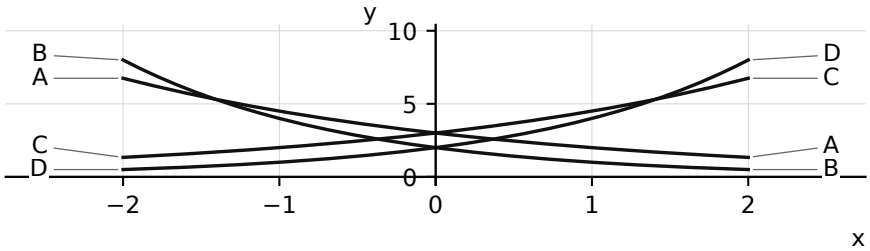
MATH 101 **92859**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-3)+3$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
2. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 3(\frac{3}{2})^x$ $(b)\quad 3(\frac{2}{3})^x$ $(c)\quad 2(2)^x$ $(d)\quad 2(\frac{1}{2})^x$ 	2. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts
3. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.4)+\log(2500)$ $(b)\quad \frac{\log_5(64)}{\log_5(16)}$	3. (a) 10 pts (b) 10 pts

Test continues on the other side of the sheet.

MATH 101 **92859**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x + y = 7$ $x - y = 1$ $(b)\quad x^2 + y^2 = 17$ $y = x + 3$ $(c)\quad x^2 + y^2 \le 17,\ y > x + 3$	4. (a) 6 pts (b) 10 pts (c) 4 pts
5. Solve both equations exactly. $(a)\quad 4^{x-3} = 2^{x+4}$ $(b)\quad \log_4(x-3) + \log_4(5) = 2$	5. (a) 8 pts (b) 12 pts

92859

TEST PAGE 1

1. (a)

$(3, \infty)$ — 4 pts

(b)

$(-\infty, \infty)$ — 4 pts

(c)

$x = 3$ — 3 pts

(d)

$(\frac{82}{27}, 0)$ — 3 pts

(e)

$x \to 3^+ : f(x) \to -\infty$ — 3 pts

(f)

$x \to \infty : f(x) \to \infty$ — 3 pts

2. (a)

C — 5 pts

(b)

A — 5 pts

(c)

D — 5 pts

(d)

B — 5 pts

3. (a)

3 — 10 pts

(b)

$\frac{3}{2}$ — 10 pts

TEST PAGE 2

4. (a)

$(4, 3)$ — 6 pts

(b)

$(1, 4); (-4, -1)$ — 10 pts

(c)

No — 4 pts

5. (a)

$x = 10$ — 8 pts

(b)

$x = \frac{31}{5}$ — 12 pts

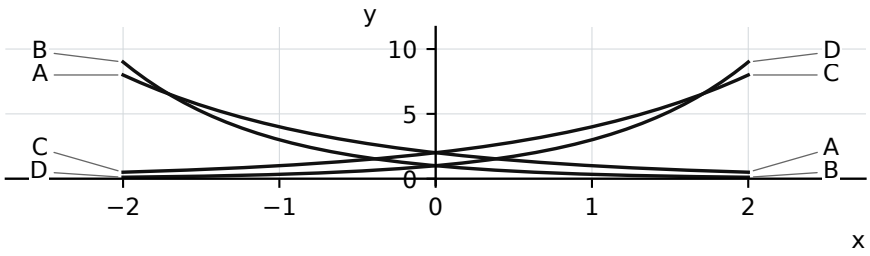
MATH 101 **33415**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=7$ $x-y=1$ $(b)\quad x^2+y^2=17$ $y=x+3$ $(c)\quad x^2+y^2\le 17,\ y>x+3$	1. (a) 6 pts (b) 10 pts (c) 4 pts
2. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(2)^x$ $(b)\quad 2(\frac{1}{2})^x$ $(c)\quad 1(3)^x$ $(d)\quad 1(\frac{1}{3})^x$ 	2. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts

Test continues on the other side of the sheet.

MATH 101 **33415**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

3. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-3) + 3$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
4. Evaluate exactly. Here log means base 10. (a) $\log(0.5) + \log(20000)$ (b) $\frac{\log_5(64)}{\log_5(16)}$	4. (a) 10 pts (b) 10 pts
5. Solve both equations exactly. (a) $4^{x-3} = 2^{x+4}$ (b) $\log_4(x-3) + \log_4(5) = 2$	5. (a) 8 pts (b) 12 pts

33415

TEST PAGE 1

1. (a)

$(4, 3)$

6 pts

(b)

$(1, 4); (-4, -1)$

10 pts

(c)

No

4 pts

2. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

TEST PAGE 2

3. (a)

$(3, \infty)$

4 pts

(b)

$(-\infty, \infty)$

4 pts

(c)

$x = 3$

3 pts

(d)

$(\frac{193}{64}, 0)$

3 pts

(e)

$x \to 3^+ : f(x) \to -\infty$

3 pts

(f)

$x \to \infty : f(x) \to \infty$

3 pts

4. (a)

4

10 pts

(b)

$\frac{3}{2}$

10 pts

5. (a)

$x = 10$

8 pts

(b)

$x = \frac{31}{5}$

12 pts

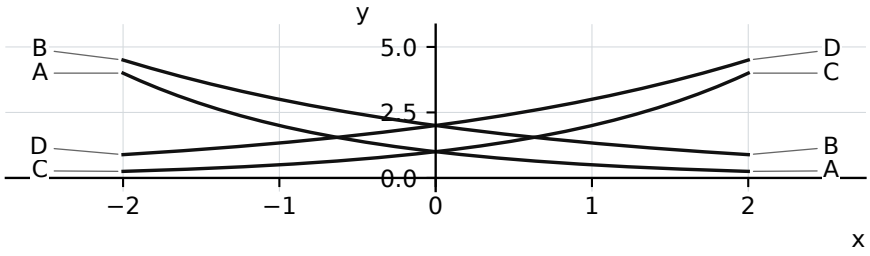
MATH 101 **64877**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=9$ $x-y=1$ $(b)\quad x^2+y^2=26$ $y=x+4$ $(c)\quad x^2+y^2\le 26,\ y>x+4$	1. (a) 6 pts (b) 10 pts (c) 4 pts
2. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 1(2)^x$ $(b)\quad 1(\frac{1}{2})^x$ $(c)\quad 2(\frac{3}{2})^x$ $(d)\quad 2(\frac{2}{3})^x$ 	2. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts

Test continues on the other side of the sheet.

MATH 101 **64877**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

3. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x-4) + 1$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
4. Evaluate exactly. Here log means base 10. (a) $\log(0.2) + \log(500)$ (b) $\frac{\log_6(125)}{\log_6(25)}$	4. (a) 10 pts (b) 10 pts
5. Solve both equations exactly. (a) $4^{x-4} = 2^{x+5}$ (b) $\log_5(x-4) + \log_5(6) = 2$	5. (a) 8 pts (b) 12 pts

64877

TEST PAGE 1

1. (a)

$(5, 4)$

6 pts

(b)

$(1, 5); (-5, -1)$

10 pts

(c)

No

4 pts

2. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

TEST PAGE 2

3. (a)

$(4, \infty)$ 4 pts

(b)

$(-\infty, \infty)$ 4 pts

(c)

$x = 4$ 3 pts

(d)

$(\frac{9}{2}, 0)$ 3 pts

(e)

$x \to 4^+ : f(x) \to -\infty$ 3 pts

(f)

$x \to \infty : f(x) \to \infty$ 3 pts

4. (a)

2 10 pts

(b)

$\frac{3}{2}$ 10 pts

5. (a)

$x = 13$ 8 pts

(b)

$x = \frac{49}{6}$ 12 pts

MATH 101 **34553**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=9$ $x-y=1$ $(b)\quad x^2+y^2=26$ $y=x+4$ $(c)\quad x^2+y^2\le 26,\ y>x+4$	1. (a) 6 pts (b) 10 pts (c) 4 pts
2. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x)=\log_3(x-4)+1$	2. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
3. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.4)+\log(2500)$ $(b)\quad \frac{\log_6(125)}{\log_6(25)}$	3. (a) 10 pts (b) 10 pts

Test continues on the other side of the sheet.

MATH 101 **34553**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

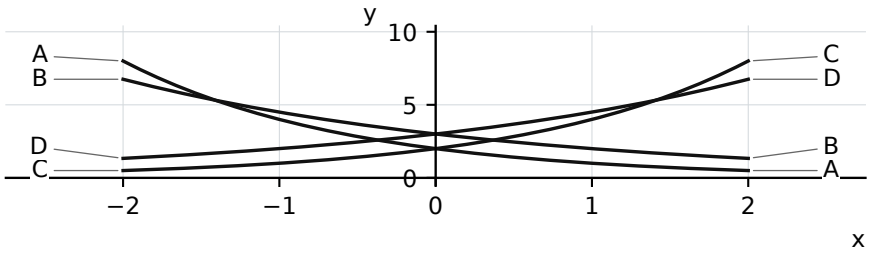
4. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $2(2)^x$

(b) $2(\frac{1}{2})^x$

(c) $3(\frac{3}{2})^x$

(d) $3(\frac{2}{3})^x$



4. (a)

5 pts

(b)

5 pts

(c)

5 pts

(d)

5 pts

5. Solve both equations exactly.

(a) $4^{x-4} = 2^{x+5}$

(b) $\log_5(x-4) + \log_5(6) = 2$

5. (a)

8 pts

(b)

12 pts

34553

TEST PAGE 1

1. (a)

$(5, 4)$

6 pts

(b)

$(1, 5); (-5, -1)$

10 pts

(c)

No

4 pts

2. (a)

$(4, \infty)$ 4 pts

(b)

$(-\infty, \infty)$ 4 pts

(c)

$x = 4$ 3 pts

(d)

$(\frac{13}{3}, 0)$ 3 pts

(e)

$x \to 4^+ : f(x) \to -\infty$ 3 pts

(f)

$x \to \infty : f(x) \to \infty$ 3 pts

3. (a)

3 10 pts

(b)

$\frac{3}{2}$ 10 pts

TEST PAGE 2

4. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

5. (a)

$x = 13$ 8 pts

(b)

$x = \frac{49}{6}$ 12 pts

MATH 101 **13446**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve both equations exactly. $(a)\quad 4^{x-4} = 2^{x+5}$ $(b)\quad \log_5(x-4) + \log_5(6) = 2$	1. (a) 8 pts (b) 12 pts
2. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-4) + 1$	2. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
3. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x + y = 9$ $x - y = 1$ $(b)\quad x^2 + y^2 = 26$ $y = x + 4$ $(c)\quad x^2 + y^2 \le 26,\ y > x + 4$	3. (a) 6 pts (b) 10 pts (c) 4 pts
4. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.5) + \log(20000)$ $(b)\quad \frac{\log_6(125)}{\log_6(25)}$	4. (a) 10 pts (b) 10 pts

Test continues on the other side of the sheet.

MATH 101 **13446**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

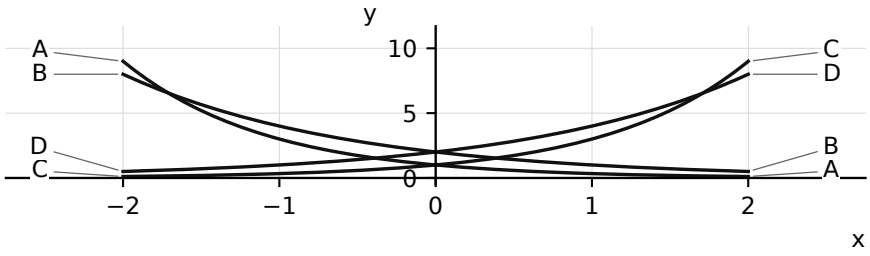
5. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $1(3)^x$

(b) $1(\frac{1}{3})^x$

(c) $2(2)^x$

(d) $2(\frac{1}{2})^x$



5. (a)

5 pts

(b)

5 pts

(c)

5 pts

(d)

5 pts

13446

TEST PAGE 1

1. (a)

$x = 13$ 8 pts

(b)

$x = \frac{49}{6}$ 12 pts

2. (a)

$(4, \infty)$ 4 pts

(b)

$(-\infty, \infty)$ 4 pts

(c)

$x = 4$ 3 pts

(d)

$(\frac{17}{4}, 0)$ 3 pts

(e)

$x \to 4^+ : f(x) \to -\infty$ 3 pts

(f)

$x \to \infty : f(x) \to \infty$ 3 pts

3. (a)

$(5, 4)$ 6 pts

(b)

$(1, 5); (-5, -1)$ 10 pts

(c)

No 4 pts

4. (a)

4 10 pts

(b)

$\frac{3}{2}$ 10 pts

TEST PAGE 2

5. (a)

C 5 pts

(b)

A 5 pts

(c)

D 5 pts

(d)

B 5 pts

MATH 101 **92601**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x-5) + 2$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
2. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a) \quad x + y = 11$ $x - y = 1$ $(b) \quad x^2 + y^2 = 37$ $y = x + 5$ $(c) \quad x^2 + y^2 \le 37,\ y > x + 5$	2. (a) 6 pts (b) 10 pts (c) 4 pts

Test continues on the other side of the sheet.

MATH 101 **92601**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

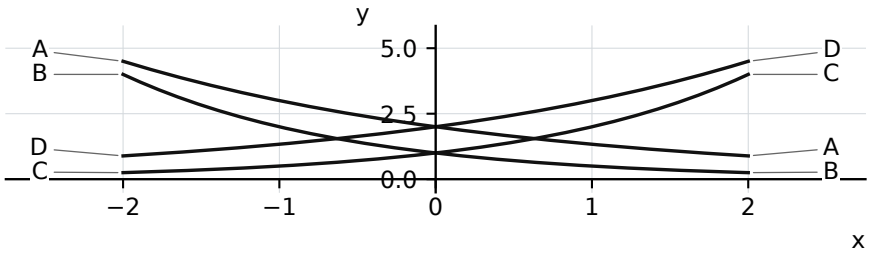
3. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $1(2)^x$

(b) $2(\frac{2}{3})^x$

(c) $2(\frac{3}{2})^x$

(d) $1(\frac{1}{2})^x$



3. (a) 5 pts

(b) 5 pts

(c) 5 pts

(d) 5 pts

4. Solve both equations exactly.

(a) $4^{x-5} = 2^{x+6}$

(b) $\log_6(x-5) + \log_6(7) = 2$

4. (a) 8 pts

(b) 12 pts

5. Evaluate exactly. Here log means base 10.

(a) $\log(0.2) + \log(500)$

(b) $\frac{\log_7(216)}{\log_7(36)}$

5. (a) 10 pts

(b) 10 pts

92601

TEST PAGE 1

1. (a)

$(5, \infty)$ 4 pts

(b)

$(-\infty, \infty)$ 4 pts

(c)

$x = 5$ 3 pts

(d)

$(\frac{21}{4}, 0)$ 3 pts

(e)

$x \to 5^+ : f(x) \to -\infty$ 3 pts

(f)

$x \to \infty : f(x) \to \infty$ 3 pts

2. (a)

$(6, 5)$ 6 pts

(b)

$(1, 6); (-6, -1)$ 10 pts

(c)

No 4 pts

TEST PAGE 2

3. (a)

C 5 pts

(b)

A 5 pts

(c)

D 5 pts

(d)

B 5 pts

4. (a)

$x = 16$ 8 pts

(b)

$x = \frac{71}{7}$ 12 pts

5. (a)

2 10 pts

(b)

$\frac{3}{2}$ 10 pts

MATH 101 **82960**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate exactly. Here log means base 10. (a) $\log(0.4) + \log(2500)$ (b) $\frac{\log_7(216)}{\log_7(36)}$	1. (a) 10 pts (b) 10 pts
2. Solve both equations exactly. (a) $4^{x-5} = 2^{x+6}$ (b) $\log_6(x-5) + \log_6(7) = 2$	2. (a) 8 pts (b) 12 pts
3. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-5) + 2$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts

Test continues on the other side of the sheet.

MATH 101 **82960**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

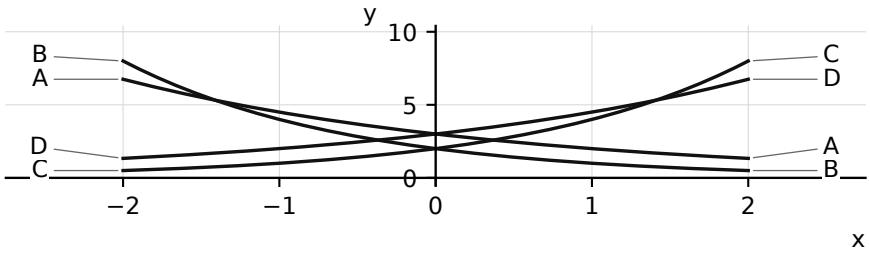
4. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 2(2)^x$

$(b)\quad 3(\frac{2}{3})^x$

$(c)\quad 3(\frac{3}{2})^x$

$(d)\quad 2(\frac{1}{2})^x$



4. (a) 5 pts

(b) 5 pts

(c) 5 pts

(d) 5 pts

5. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership.

$(a)\quad x + y = 11$

$x - y = 1$

$(b)\quad x^2 + y^2 = 37$

$y = x + 5$

$(c)\quad x^2 + y^2 \leq 37,\ y > x + 5$

5. (a) 6 pts

(b) 10 pts

(c) 4 pts

82960

TEST PAGE 1

1. (a)

3 — 10 pts

(b)

$\frac{3}{2}$ — 10 pts

2. (a)

$x = 16$ — 8 pts

(b)

$x = \frac{71}{7}$ — 12 pts

3. (a)

$(5, \infty)$ — 4 pts

(b)

$(-\infty, \infty)$ — 4 pts

(c)

$x = 5$ — 3 pts

(d)

$(\frac{46}{9}, 0)$ — 3 pts

(e)

$x \to 5^+ : f(x) \to -\infty$ — 3 pts

(f)

$x \to \infty : f(x) \to \infty$ — 3 pts

TEST PAGE 2

4. (a)

C — 5 pts

(b)

A — 5 pts

(c)

D — 5 pts

(d)

B — 5 pts

5. (a)

$(6, 5)$ — 6 pts

(b)

$(1, 6); (-6, -1)$ — 10 pts

(c)

No — 4 pts

MATH 101 **41079**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

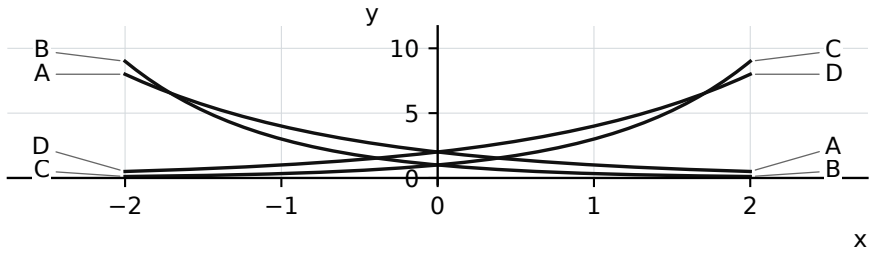
1. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $1(3)^x$

(b) $2(\frac{1}{2})^x$

(c) $2(2)^x$

(d) $1(\frac{1}{3})^x$



1. (a) 5 pts

(b) 5 pts

(c) 5 pts

(d) 5 pts

2. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_4(x-5)+2$

2. (a) 4 pts

(b) 4 pts

(c) 3 pts

(d) 3 pts

(e) 3 pts

(f) 3 pts

3. Solve both equations exactly.

(a) $4^{x-5} = 2^{x+6}$

(b) $\log_6(x-5)+\log_6(7)=2$

3. (a) 8 pts

(b) 12 pts

Test continues on the other side of the sheet.

MATH 101 **41079**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.5)+\log(20000)$ $(b)\quad \frac{\log_7(216)}{\log_7(36)}$	4. (a) 10 pts (b) 10 pts
5. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=11$ $x-y=1$ $(b)\quad x^2+y^2=37$ $y=x+5$ $(c)\quad x^2+y^2\le 37,\ y>x+5$	5. (a) 6 pts (b) 10 pts (c) 4 pts

41079

TEST PAGE 1

1. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

2. (a)

$(5, \infty)$

4 pts

(b)

$(-\infty, \infty)$

4 pts

(c)

$x = 5$

3 pts

(d)

$(\frac{81}{16}, 0)$

3 pts

(e)

$x \to 5^+ : f(x) \to -\infty$

3 pts

(f)

$x \to \infty : f(x) \to \infty$

3 pts

3. (a)

$x = 16$

8 pts

(b)

$x = \frac{71}{7}$

12 pts

TEST PAGE 2

4. (a)

4

10 pts

(b)

$\frac{3}{2}$

10 pts

5. (a)

$(6, 5)$

6 pts

(b)

$(1, 6); (-6, -1)$

10 pts

(c)

No

4 pts

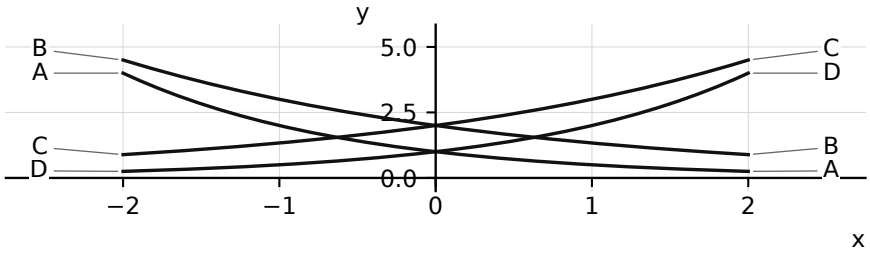
MATH 101 **84906**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=13$ $x-y=1$ $(b)\quad x^2+y^2=50$ $y=x+6$ $(c)\quad x^2+y^2\le 50,\ y>x+6$	1. (a) 6 pts (b) 10 pts (c) 4 pts
2. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(\frac{3}{2})^x$ $(b)\quad 1(\frac{1}{2})^x$ $(c)\quad 1(2)^x$ $(d)\quad 2(\frac{2}{3})^x$ 	2. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts

Test continues on the other side of the sheet.

MATH 101 **84906**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

3. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_2(x-6) + 3$	3. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
4. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.2) + \log(500)$ $(b)\quad \frac{\log_8(343)}{\log_8(49)}$	4. (a) 10 pts (b) 10 pts
5. Solve both equations exactly. $(a)\quad 4^{x-6} = 2^{x+7}$ $(b)\quad \log_7(x-6) + \log_7(8) = 2$	5. (a) 8 pts (b) 12 pts

84906

TEST PAGE 1

1. (a)

$(7, 6)$

6 pts

(b)

$(1, 7); (-7, -1)$

10 pts

(c)

No

4 pts

2. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

TEST PAGE 2

3. (a)

$(6, \infty)$

4 pts

(b)

$(-\infty, \infty)$

4 pts

(c)

$x = 6$

3 pts

(d)

$(\frac{49}{8}, 0)$

3 pts

(e)

$x \to 6^+ : f(x) \to -\infty$

3 pts

(f)

$x \to \infty : f(x) \to \infty$

3 pts

4. (a)

2

10 pts

(b)

$\frac{3}{2}$

10 pts

5. (a)

$x = 19$

8 pts

(b)

$x = \frac{97}{8}$

12 pts

MATH 101 **87610**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x-6) + 3$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
2. Solve both equations exactly. $(a)\quad 4^{x-6} = 2^{x+7}$ $(b)\quad \log_7(x-6) + \log_7(8) = 2$	2. (a) 8 pts (b) 12 pts
3. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x + y = 13$ $x - y = 1$ $(b)\quad x^2 + y^2 = 50$ $y = x + 6$ $(c)\quad x^2 + y^2 \le 50,\ y > x + 6$	3. (a) 6 pts (b) 10 pts (c) 4 pts
4. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.4) + \log(2500)$ $(b)\quad \frac{\log_8(343)}{\log_8(49)}$	4. (a) 10 pts (b) 10 pts

Test continues on the other side of the sheet.

MATH 101 **87610**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

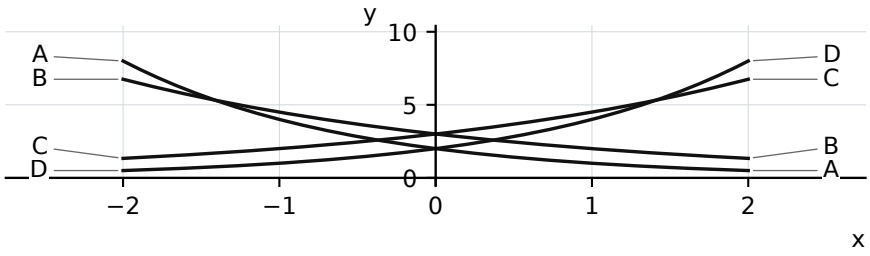
5. Match each function to graph A, B, C, or D. Each letter is used once.

(*a*) $3(\frac{3}{2})^x$

(*b*) $2(\frac{1}{2})^x$

(*c*) $2(2)^x$

(*d*) $3(\frac{2}{3})^x$



5. (a)

5 pts

(b)

5 pts

(c)

5 pts

(d)

5 pts

87610

TEST PAGE 1

1. (a)

$(6, \infty)$ 4 pts

(b)

$(-\infty, \infty)$ 4 pts

(c)

$x = 6$ 3 pts

(d)

$(\frac{163}{27}, 0)$ 3 pts

(e)

$x \to 6^+ : f(x) \to -\infty$ 3 pts

(f)

$x \to \infty : f(x) \to \infty$ 3 pts

2. (a)

$x = 19$ 8 pts

(b)

$x = \frac{97}{8}$ 12 pts

3. (a)

$(7, 6)$ 6 pts

(b)

$(1, 7); (-7, -1)$ 10 pts

(c)

No 4 pts

4. (a)

3 10 pts

(b)

$\frac{3}{2}$ 10 pts

TEST PAGE 2

5. (a)

C 5 pts

(b)

A 5 pts

(c)

D 5 pts

(d)

B 5 pts

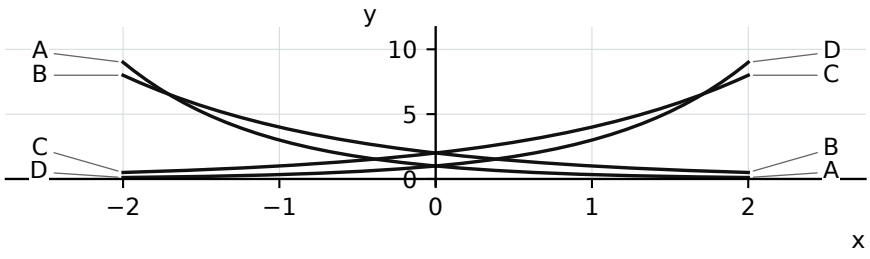
MATH 101 **18817**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_4(x-6)+3$	1. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
2. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.5)+\log(20000)$ $(b)\quad \frac{\log_8(343)}{\log_8(49)}$	2. (a) 10 pts (b) 10 pts
3. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(2)^x$ $(b)\quad 1(\frac{1}{3})^x$ $(c)\quad 1(3)^x$ $(d)\quad 2(\frac{1}{2})^x$ 	3. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts

Test continues on the other side of the sheet.

MATH 101 **18817**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Solve both equations exactly. $(a)\quad 4^{x-6} = 2^{x+7}$ $(b)\quad \log_7(x-6) + \log_7(8) = 2$	4. (a) 8 pts (b) 12 pts
5. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x + y = 13$ $x - y = 1$ $(b)\quad x^2 + y^2 = 50$ $y = x + 6$ $(c)\quad x^2 + y^2 \le 50,\ y > x + 6$	5. (a) 6 pts (b) 10 pts (c) 4 pts

18817

TEST PAGE 1

1. (a)
$(6, \infty)$ — 4 pts

(b)
$(-\infty, \infty)$ — 4 pts

(c)
$x = 6$ — 3 pts

(d)
$(\frac{385}{64}, 0)$ — 3 pts

(e)
$x \to 6^+ : f(x) \to -\infty$ — 3 pts

(f)
$x \to \infty : f(x) \to \infty$ — 3 pts

2. (a)
4 — 10 pts

(b)
$\frac{3}{2}$ — 10 pts

3. (a)
C — 5 pts

(b)
A — 5 pts

(c)
D — 5 pts

(d)
B — 5 pts

TEST PAGE 2

4. (a)
$x = 19$ — 8 pts

(b)
$x = \frac{97}{8}$ — 12 pts

5. (a)
$(7, 6)$ — 6 pts

(b)
$(1, 7); (-7, -1)$ — 10 pts

(c)
No — 4 pts

MATH 101 **57424**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

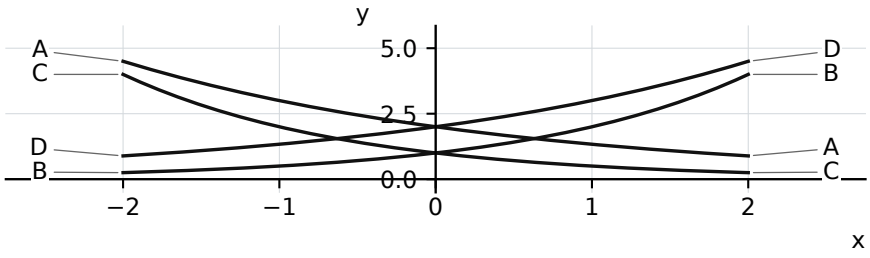
1. Match each function to graph A, B, C, or D. Each letter is used once.

(a) $1(\frac{1}{2})^x$

(b) $2(\frac{2}{3})^x$

(c) $2(\frac{3}{2})^x$

(d) $1(2)^x$



1. (a) 5 pts

(b) 5 pts

(c) 5 pts

(d) 5 pts

2. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end.

$f(x) = \log_2(x - 7) + 1$

2. (a) 4 pts

(b) 4 pts

(c) 3 pts

(d) 3 pts

(e) 3 pts

(f) 3 pts

3. Evaluate exactly. Here log means base 10.

(a) $\log(0.2) + \log(500)$

(b) $\frac{\log_9(512)}{\log_9(64)}$

3. (a) 10 pts

(b) 10 pts

Test continues on the other side of the sheet.

MATH 101 **57424**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=15$ $x-y=1$ $(b)\quad x^2+y^2=65$ $y=x+7$ $(c)\quad x^2+y^2\le 65,\ y>x+7$	4. (a) 6 pts (b) 10 pts (c) 4 pts
5. Solve both equations exactly. $(a)\quad 4^{x-7}=2^{x+8}$ $(b)\quad \log_8(x-7)+\log_8(9)=2$	5. (a) 8 pts (b) 12 pts

57424

TEST PAGE 1

1. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

2. (a)

$(7, \infty)$

4 pts

(b)

$(-\infty, \infty)$

4 pts

(c)

$x = 7$

3 pts

(d)

$(\frac{15}{2}, 0)$

3 pts

(e)

$x \to 7^+ : f(x) \to -\infty$

3 pts

(f)

$x \to \infty : f(x) \to \infty$

3 pts

3. (a)

2

10 pts

(b)

$\frac{3}{2}$

10 pts

TEST PAGE 2

4. (a)

$(8, 7)$

6 pts

(b)

$(1, 8); (-8, -1)$

10 pts

(c)

No

4 pts

5. (a)

$x = 22$

8 pts

(b)

$x = \frac{127}{9}$

12 pts

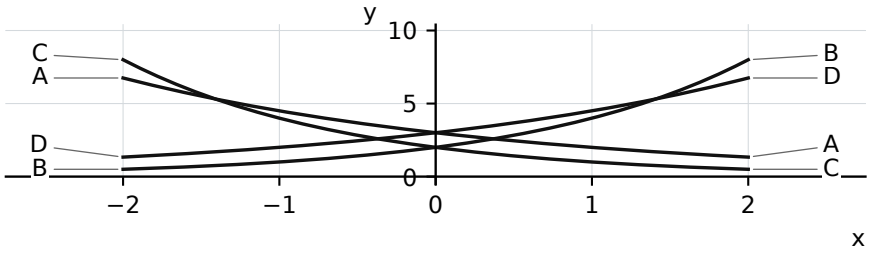
MATH 101 **60303**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the linear system. Solve the nonlinear system. Test whether (0,0) belongs to the region given by the inequalities. Record: (a) ordered pair; (b) intersections; (c) membership. $(a)\quad x+y=15$ $x-y=1$ $(b)\quad x^2+y^2=65$ $y=x+7$ $(c)\quad x^2+y^2\le 65,\ y>x+7$	1. (a) 6 pts (b) 10 pts (c) 4 pts
2. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 2(\frac{1}{2})^x$ $(b)\quad 3(\frac{2}{3})^x$ $(c)\quad 3(\frac{3}{2})^x$ $(d)\quad 2(2)^x$ 	2. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts
3. Evaluate exactly. Here log means base 10. $(a)\quad \log(0.4)+\log(2500)$ $(b)\quad \frac{\log_9(512)}{\log_9(64)}$	3. (a) 10 pts (b) 10 pts

Test continues on the other side of the sheet.

MATH 101 **60303**

Unit 4: Exponentials, Logarithms, and Systems

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find domain and range; vertical asymptote and x-intercept; behavior at both ends of the domain. Record: (a) domain; (b) range; (c) asymptote; (d) x-intercept; (e) left end; (f) right end. $f(x) = \log_3(x - 7) + 1$	4. (a) 4 pts (b) 4 pts (c) 3 pts (d) 3 pts (e) 3 pts (f) 3 pts
5. Solve both equations exactly. $(a)\quad 4^{x-7} = 2^{x+8}$ $(b)\quad \log_8(x - 7) + \log_8(9) = 2$	5. (a) 8 pts (b) 12 pts

60303

TEST PAGE 1

1. (a)

$(8, 7)$

6 pts

(b)

$(1, 8); (-8, -1)$

10 pts

(c)

No

4 pts

2. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

3. (a)

3

10 pts

(b)

$\frac{3}{2}$

10 pts

TEST PAGE 2

4. (a)

$(7, \infty)$

4 pts

(b)

$(-\infty, \infty)$

4 pts

(c)

$x = 7$

3 pts

(d)

$(\frac{22}{3}, 0)$

3 pts

(e)

$x \to 7^+ : f(x) \to -\infty$

3 pts

(f)

$x \to \infty : f(x) \to \infty$

3 pts

5. (a)

$x = 22$

8 pts

(b)

$x = \frac{127}{9}$

12 pts